8.012 Problem Set 6

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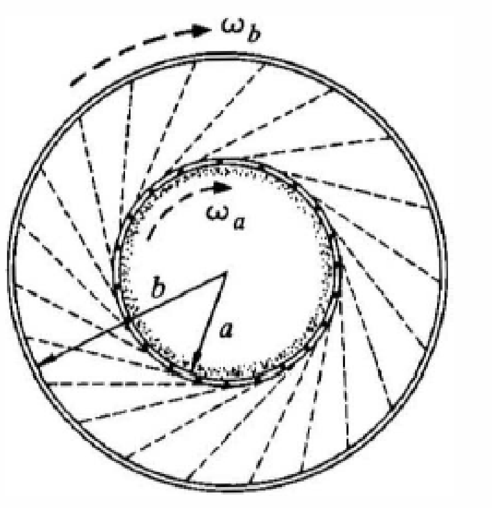
Problem 4

Kleppner & Kolenkow, Problem 6.2

6.2 A drum of mass M_A and radius a rotates freely with initial angular velocity $\omega_A(0)$. A second drum with mass M_B and radius $b > a$ is mounted on the same axis and is at rest, although it is free to rotate. A thin layer of sand with mass M_S is distributed on the inner surface of the smaller drum. At $t = 0$, small perforations in the inner drum are opened. The sand starts to fly out at a constant rate λ and sticks to the outer drum. Find the subsequent angular velocities of the two drums ω_A and ω_B. Ignore the transit time of the sand.

Ans. clue. If $\lambda t = M_b$ and $b = 2a$, then $\omega_B = \omega_A(0)/8$



Note that there are no External Torque applied on this System. So we can use Conservation of Angular Momentum.

Therefore,

$$(M_A+M_S)a^2\omega_A(0)+(M_B)b^2\omega_B(0) = (M_A+M_S-\lambda\,t)a^2\omega_A(t)+(M_B+\lambda\,t)b^2\omega_B(t)$$

$$\implies (M_B+\lambda\,t)b^2\omega_B(t) = (M_A+M_S)a^2\omega_A(0) - (M_A+M_S-\lambda\,t)a^2\omega_A(t)$$

In this problem there is Angular Momentum carried over by the Sand from the Inner to the Outer Drum.
Say the Angular momentum transported by the sand from the inner drum to the outer drum in time t be L_S

We get them as

$$L_S = (\text{Mass of sand removed})$$
$$(\text{Tangential Speed at which the Sand moves away})$$
$$(\text{Perpendicular Length of the Trajectory of sand from the Axis})$$

$$\implies L_S = (\lambda\,t)(\omega_A(t)a)(a) = a^2\lambda\omega_A(t)t$$

This Gives us

$$(M_B+\lambda\,t)b^2\omega_B(t) = a^2\lambda\omega_A(t)t + (M_B)b^2\omega_B(0) = a^2\lambda\omega_A(t)t$$

$$\implies (M_A+M_S)a^2\omega_A(0) - (M_A+M_S-\lambda\,t)a^2\omega_A(t) = a^2\lambda\omega_A(t)t$$

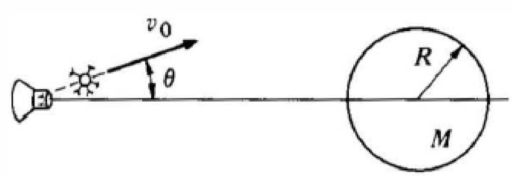
Hence,

$$\omega_A(t) = \omega_A(0)$$
$$\omega_B(t) = \frac{a^2\lambda\omega_A(0)t}{b^2(M_B+\lambda\,t)}$$

Problem 5

Kleppner & Kolenkow, Problem 6.4

6.4 A spaceship is sent to investigate a planet of mass M and radius R. While hanging motionless in space at a distance $5R$ from the center of the planet, the ship fires an instrument package with speed v_0, as shown in the sketch. The package has mass m, which is much smaller than the mass of the spaceship. For what angle θ will the package just graze the surface of the planet?



Let v be the Velocity of the Package when it grazes the Surface. By Conservation of Energy

$$\frac{mv_0^2}{2} - \frac{GMm}{5R} = \frac{mv^2}{2} - \frac{GMm}{R} \implies v = \sqrt{v_0^2 + \frac{8GM}{5R}}$$

Gravitational Force is a Central Force and there is no Tangential Force acting on the Package. So, Angular Momentum is Conserved. Using Conservation of Angular Momentum of the Package about the Center of the Planet gives us

$$mv_0 \sin(\theta) 5R = mvR \implies \sin(\theta) = \frac{v}{5v_0}$$

Hence,

$$\theta = \arcsin\left(\frac{1}{5v_0}\sqrt{v_0^2 + \frac{8GM}{5R}}\right)$$